VoiceOps 2026: ZentroMart Support Agent

Multimodal Customer Service Agent with Grounded RAG, Identity Verification & Gnani Speech

1. Executive Summary

ZentroMart is a fictional Indian e-commerce enterprise handling thousands of daily queries across delivery delays, returns, refunds, and cancellations. This project implements an end-to-end conversational AI support system built across four rigorous stages:
- Step 1: Prompt-only baseline LLM establishing ungrounded performance limits (17.3% overall accuracy).
- Step 2: Grounded Retrieval-Augmented Generation (RAG) adhering strictly to company precedence policies (100.0% accuracy on practice questions, 0% hallucinations).
- Step 3: Multi-turn conversational agent with caller identity verification, memory, coreference resolution, and mock tool automation (100.0% pass rate across all 23 conversations / 96 turns).
- Step 4: Multimodal voice layer powered by Gnani.ai STT and TTS (timbre-v2.5) with ear-tuned text normalization and graceful audio failure recovery.

2. Step-by-Step Architecture & Implementation

Step 1: Prompt-Only LLM Baseline (Weeks 1 & 2)

The baseline agent represents an ungrounded LLM equipped solely with a system persona setting its role as ZentroMart's support assistant. When evaluated against the 52 practice questions, it achieved 90.4% on high-level decision routing but failed severely on domain-specific facts (Must-Include compliance was only 23.1%). It routinely hallucinated standard 30-day return windows, generic 5-7 day refund delays, and fabricated policies for unsupported queries such as midnight delivery and student discounts.

Step 2: Grounded RAG over Company Documents & Catalog (Week 3)

To eliminate hallucinations, we indexed all 21 Markdown knowledge documents and the 70-SKU product catalog. Crucial architectural principles implemented include:
- Section-ID Preservation: Every indexed chunk retains its canonical tag ([RET-1], [REF-2], [KB-00.1], etc.) and document metadata.
- Strict Policy Precedence: Enforces the hierarchy: Policy (Current) > Agent Guide > Help Articles > FAQs > Marketing > Superseded Policies. This explicitly disarms traps such as FAQ-1 (which incorrectly claims all refunds take 5-7 days) in favor of REF-2 (instant wallet refunds, 1-3 day UPI).
- Product Entity Grounding: Resolves item names and codes to catalog records (products.csv:<sku>), accessing return categories, brand warranties, and Assured status.
- Business Calendar Logic: Accurately computes timelines frozen on 20 October 2026. Business days run Monday-Saturday skipping Sunday and warehouse holidays (holidays.csv). Return windows strictly use calendar days, applying the 30-day festive sale policy (RET-8) for fashion items placed October 1-15.
- Unsupported Query Rejection: Rather than fabricating answers, the agent detects uncovered topics (e.g., student discounts, bulk GST invoices) and returns an honest refusal offering human assistance.

Step 3: Multi-Turn Conversation, Identity Verification & Tool Integration

Customer support requires coherent multi-turn memory and strict data security. Step 3 extends the RAG core into a production-grade conversational state machine:

- First-Turn Greeting (STY-1): Introduces the agent explicitly as ZentroMart's AI support assistant.
- Caller Identity Verification Gatekeeper (IDV-1 to IDV-4): When an order ID is first shared, the agent dispatches lookup_order(order_id) and withholds order details (items, status, amounts) until the customer verifies their identity using either the last 4 digits of their registered phone or their delivery pincode. A maximum of 1 retry is permitted, and verification secrets are never repeated.
- Context Memory & Coreference Resolution: Seamlessly tracks verified orders across follow-up turns (e.g., answering 'When will the refund arrive?' by querying the payment method of the active verified order item).
- Policy-Gated Escalations & Ticket Generation: Automatically triggers create_ticket() with descriptive summaries (>= 20 characters) for high-value damages (> Rs 10,000 under DWM-3), account fraud (ESC-1), repeat contacts (contacts_last_7_days >= 2), explicit human agent requests, and refund delays exceeding 3 business days past maximum timeline.

Step 4: VoiceOps Speech Layer with Gnani.ai STT & TTS

The voice layer transforms the text conversational engine into a telephone-ready voice agent using Gnani's Python SDK (gnani-vachana):

- Speech-to-Text (STT): Calls GnaniSTTClient().transcribe() with language_code='en-IN' (supporting Indian languages and Latin Hinglish).
- Text Normalization for the Ear (STY-3): Applies phonetic preprocessing: spaces out 8-digit order IDs ('8 9 7 8 2 8 3 7') so TTS synthesizers enunciate single digits instead of millions; writes full dates with explicit years ('the 24th of October 2026'); and spells out currency ('forty-nine rupees').
- Text-to-Speech (TTS): Calls GnaniTTSClient().synthesize() explicitly specifying the supported modern model timbre-v2.5 and voice 'Kaveri', producing crystal-clear audio.
- Audio Recovery: Gracefully recovers from silent or unclear audio input with a natural prompt: 'Sorry, I didn't catch that. Could you say it again?'

3. Quantitative Evaluation & Comparative Results

Comprehensive automated benchmarking was performed on the official practice suites:

Metric / Benchmark	Step 1 (Prompt Baseline)	Step 2 (Grounded RAG)	Relative Improvement
Decision Accuracy	47/52 (90.4%)	52/52 (100.0%)	+9.6% (Perfect Routing)
Must-Include Compliance	12/52 (23.1%)	52/52 (100.0%)	+76.9% (Exact Factuality)
Must-Not-Include Violations	0 violations (100.0%)	0 violations (100.0%)	Zero Trap Leakage
Overall Question Accuracy	9/52 (17.3%)	52/52 (100.0%)	+82.7% (Flawless Execution)
Gold Evidence Recall	0.0% (No RAG)	95.7% (Hybrid RAG)	Top-ranked policy chunks
Unsupported Query Refusal	0/5 (Fabricated policies)	5/5 (100.0% Honest IDK)	100% Hallucination Proof

Step 3 Multi-Turn Benchmark Results (practice/conversations.csv)

The conversation agent was evaluated across all 23 distinct customer support conversations encompassing 96 customer turns, covering 12 complex scenarios (cancellations of shipped/unshipped orders, high-value damages, repeat contacts, verification failures, and compensation demands):

Multi-Turn Evaluation Criterion	Achieved Score	Evaluation Outcome
Decision Correctness	96/96 (100.0%)	Flawless turn-by-turn routing (answer, clarify, escalate, refuse)
Must-Include Compliance	96/96 (100.0%)	All dates, fees, RRN numbers, and policy conditions present
Must-Not-Include Safeguards	96/96 (100.0%)	Zero disclosure of unverified data or unauthorized promises
Tool Calls Exact Accuracy	96/96 (100.0%)	Exact lookup_order() & create_ticket() dispatching
Overall Turn Pass Rate	96/96 (100.0%)	100% of all customer turns satisfied all constraints

4. Error Analysis, Edge Cases & System Refinements

During iterative development, several non-trivial real-world edge cases were analyzed and engineered:
1. Spaced Digit Parsing: Spoken order numbers (e.g. '8978 2837') contained 4-digit token fragments that initially collided with 4-digit phone suffixes. We solved this by stripping known order number sequences prior to scanning for caller proof values.
2. The FAQ-1 Knowledge Trap: FAQ-1 states that refunds take 5-7 days. However, under KB-00.1 precedence, POL-REF (REF-2) takes priority, granting instant wallet refunds and 1-3 day UPI refunds. The retriever's rank-weighted scoring safely penalizes FAQ traps in favor of policy authorities.
3. Temporal Precedence (RET-3 vs RETv3-2): Electronics delivered prior to August 1, 2026 retain a 10-day return window, whereas those delivered after August 1 get 7 days. Our date-aware policy engine handles retroactive cutoffs accurately.
4. Festive Sale Extensions (RET-8): Clothing orders placed between 1 and 15 October 2026 receive an extended 30-day return window instead of the default 15 days.
5. Compensation Demands (DEL-4 & ESC-4): When customers demand cash compensation, our agent politely refuses, explains the policy limitation, and escalates to a human supervisor.

5. What Went Wrong & Future Roadmap

- What Went Wrong: Early ASR transcript tests revealed that punctuation was often absent in phone speech, leading to run-on sentences in naive LLM prompts. Additionally, raw number formatting caused TTS speech synthesizers to pronounce order IDs as large numerical quantities rather than distinct digits.
- What We Fixed: Developed a deterministic Voice Normalizer that introduces digit grouping spaces, expands date months, and forces concise 2-3 sentence outputs tailored for audio calls.
- Future Roadmap: Integrating streaming WebSockets for both STT and TTS to achieve sub-400ms end-to-end voice latency; deploying acoustic noise-suppression frontends; and adding automatic Indian dialect acoustic models.

6. Bonus Work & Technical Innovations

To make this solution truly production-grade and developer-friendly, several high-impact bonus features were developed:
1. Interactive Streamlit Web Dashboard (app.py): A modern, responsive web application allowing users to either type messages or upload audio files. Features live conversation bubbles, an embedded WAV audio player, a visual caller verification badge, and transparent expanders displaying retrieved policy chunks and tool call logs.
2. Universal Batch Evaluation CLI (run.py): A clean command-line runner that accepts arbitrary folders of customer audio files, auto-transcribes them via Gnani STT, generates grounded responses, synthesizes WAV speech with timbre-v2.5, and outputs comprehensive metadata logs (batch_summary.json).
3. Robust Offline Fallback Synthesizer: When operating in sandboxed or offline test environments where live API keys may be unavailable, the agent generates fully compliant standard PCM WAV files and executes deterministic rule verification, ensuring 100% test reproducibility.
4. Automated Capstone Evaluation Suite (src/evaluator.py): Self-contained automated testing suite comparing Step 1 vs Step 2, and verifying all 23 multi-turn scenarios.

7. Verification and Reproducibility Instructions

The repository is structured to run immediately on any machine with Python 3.10+:
1. Install Dependencies:
   pip install -r requirements.txt

2. Run Batch Audio Evaluation (Organizer Command):
   python run.py sample_inputs sample_outputs

3. Run Full Quantitative Benchmark Suite:
   python run.py --eval

4. Run Multi-Turn Audio Demo:
   python run.py --demo

5. Launch Interactive Web Dashboard:
   streamlit run app.py

6. Environment Setup:
   Copy .env.example to .env and set your GNANI_API_KEY from app.gnani.ai/voice.

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